



CASO DE ESTUDIO

Proyecto INF3772

CONFIGURACIÓN ROUTING



SVI – enrutamiento inter vlan

Se creo una interface virtual para cada VLAN en la capa de distribución para que pueda enrutar el tráfico con las otras VLANs de la misma manera en los switches de núcleo solamente para la VLAN 50 Servidores.

```
MS-D11# show ip interface brief
GigabitEthernet1/0/23 unassigned YES unset up up
GigabitEthernet1/0/24 unassigned YES unset down down
GigabitEthernet1/1 unassigned YES manual up up
GigabitEthernet1/1/1 172.16.15.77 YES manual up up
GigabitEthernet1/1/2 unassigned YES unset down down
GigabitEthernet1/1/3 unassigned YES unset down down
GigabitEthernet1/1/4 unassigned YES unset down down
Vlan1 unassigned YES unset administratively down down
Vlan10 172.16.0.2 YES manual up up
Vlan15 172.16.8.2 YES manual up up
Vlan20 172.16.10.2 YES manual up up
Vlan25 172.16.11.2 YES manual up up
Vlan30 172.16.12.2 YES manual up up
Vlan35 172.16.13.2 YES manual up up
Vlan40 172.16.14.2 YES manual up up
Vlan45 172.16.14.66 YES manual up up
Vlan55 172.16.14.162 YES manual up up
Vlan60 172.16.14.194 YES manual up up
Vlan65 172.16.14.226 YES manual up up
Vlan70 172.16.15.2 YES manual up up
Vlan75 172.16.15.34 YES manual up up
MS-D11# show ip interface brief

MS-D22# show ip interface brief
Vlan25 172.16.11.3 YES manual up up
Vlan30 172.16.12.3 YES manual up up
Vlan35 172.16.13.3 YES manual up up
Vlan40 172.16.14.3 YES manual up up
Vlan45 172.16.14.67 YES manual up up
Vlan55 172.16.14.163 YES manual up up
Vlan60 172.16.14.195 YES manual up up
Vlan65 172.16.14.227 YES manual up up
Vlan70 172.16.15.3 YES manual up up
Vlan75 172.16.15.35 YES manual up up
MS-D22# show ip interface brief
```

Rutas directamente conectadas en los swithes L3 de distribucion

```
MS-D11# show ip route
MS-D11# show ip route connected
C 172.16.0.0/21 is directly connected, Vlan10
C 172.16.8.0/23 is directly connected, Vlan15
C 172.16.10.0/24 is directly connected, Vlan20
C 172.16.11.0/24 is directly connected, Vlan25
C 172.16.12.0/24 is directly connected, Vlan30
C 172.16.13.0/24 is directly connected, Vlan35
C 172.16.14.0/26 is directly connected, Vlan40
C 172.16.14.64/26 is directly connected, Vlan45
C 172.16.14.160/27 is directly connected, Vlan55
C 172.16.14.192/27 is directly connected, Vlan60
C 172.16.14.224/27 is directly connected, Vlan65
C 172.16.15.0/27 is directly connected, Vlan70
C 172.16.15.32/27 is directly connected, Vlan75
C 172.16.15.76/30 is directly connected, GigabitEthernet1/1/1
MS-D11#

MS-D22# show ip route connected
C 172.16.0.0/21 is directly connected, Vlan10
C 172.16.8.0/23 is directly connected, Vlan15
C 172.16.10.0/24 is directly connected, Vlan20
C 172.16.11.0/24 is directly connected, Vlan25
C 172.16.12.0/24 is directly connected, Vlan30
C 172.16.13.0/24 is directly connected, Vlan35
C 172.16.14.0/26 is directly connected, Vlan40
C 172.16.14.64/26 is directly connected, Vlan45
C 172.16.14.160/27 is directly connected, Vlan55
C 172.16.14.192/27 is directly connected, Vlan60
C 172.16.14.224/27 is directly connected, Vlan65
C 172.16.15.0/27 is directly connected, Vlan70
C 172.16.15.32/27 is directly connected, Vlan75
C 172.16.15.80/30 is directly connected, GigabitEthernet1/1/1
MS-D22#
```

Routing OSPFv2

Todos los switches L3 y el router de borde forman parte del proceso OSPF 1 en el área 0.

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Nombre del dispositivo	Router ID
MS-D11	3.3.3.3
MS-D22	4.4.4.4
MS-C11	1.1.1.1
MS-C22	2.2.2.2
RB	6.6.6.6

```
MS-C11#show ip route ospf 1
172.16.0.0/16 is variably subnetted, 23 subnets, 7 masks
O 172.16.0.0 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.8.0 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.10.0 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.11.0 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.12.0 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.13.0 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.14.0 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.14.64 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.14.160 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.14.192 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.14.224 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.15.0 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.15.32 [110/2] via 172.16.15.77, 00:46:33, GigabitEthernet1/1/1
O 172.16.15.68 [110/2] via 172.16.15.66, 00:46:33, GigabitEthernet1/1/2
O 172.16.14.131 [110/2] via 172.16.14.131, 00:46:33, Vlan50
O 172.16.15.74 [110/2] via 172.16.15.74, 00:46:33, Port-channel2
O 172.16.15.80 [110/2] via 172.16.14.131, 00:46:33, Vlan50
O 172.16.15.74 [110/2] via 172.16.15.74, 00:46:33, Port-channel2
```

```
MS-D11#show ip route ospf 1
172.16.0.0/16 is variably subnetted, 35 subnets, 7 masks
O 172.16.14.128 [110/2] via 172.16.15.78, 00:47:17, GigabitEthernet1/1/1
O 172.16.15.64 [110/2] via 172.16.15.78, 00:47:17, GigabitEthernet1/1/1
O 172.16.15.68 [110/3] via 172.16.15.78, 00:47:17, GigabitEthernet1/1/1
O 172.16.0.3 [110/3] via 172.16.0.3, 00:47:17, Vlan10
O 172.16.8.3 [110/3] via 172.16.8.3, 00:47:17, Vlan15
O 172.16.10.3 [110/3] via 172.16.10.3, 00:47:17, Vlan20
O 172.16.11.3 [110/3] via 172.16.11.3, 00:47:17, Vlan25
O 172.16.12.3 [110/3] via 172.16.12.3, 00:47:17, Vlan30
O 172.16.13.3 [110/3] via 172.16.13.3, 00:47:17, Vlan35
O 172.16.14.3 [110/3] via 172.16.14.3, 00:47:17, Vlan40
O 172.16.14.67 [110/3] via 172.16.14.67, 00:47:17, Vlan45
O 172.16.14.163 [110/3] via 172.16.14.163, 00:47:17, Vlan55
O 172.16.14.195 [110/3] via 172.16.14.195, 00:47:17, Vlan60
O 172.16.14.227 [110/3] via 172.16.14.227, 00:47:17, Vlan65
O 172.16.15.3 [110/3] via 172.16.15.3, 00:47:17, Vlan70
O 172.16.15.35 [110/3] via 172.16.15.35, 00:47:17, Vlan75
O 172.16.15.72 [110/2] via 172.16.15.78, 00:47:17, GigabitEthernet1/1/1
O 172.16.15.80 [110/2] via 172.16.0.3, 00:47:17, Vlan10
```

El router de borde tiene una ruta por default hacia el ISP, la ruta se redistribuye a través de OSPFv2 a todos los router de la empresa.

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```
RB#show ip route
O    172.16.15.32/27 [110/3] via 172.16.15.69, 00:48:10, GigabitEthernet0/2/0
C    172.16.15.64/30 is directly connected, GigabitEthernet0/3/0
L    172.16.15.66/32 is directly connected, GigabitEthernet0/3/0
C    172.16.15.68/30 is directly connected, GigabitEthernet0/2/0
L    172.16.15.70/32 is directly connected, GigabitEthernet0/2/0
O    172.16.15.72/30 [110/2] via 172.16.15.65, 00:48:10, GigabitEthernet0/3/0
O    172.16.15.76/30 [110/2] via 172.16.15.69, 00:48:10, GigabitEthernet0/2/0
O    172.16.15.80/30 [110/2] via 172.16.15.65, 00:48:55, GigabitEthernet0/3/0
O    172.16.15.80/30 [110/2] via 172.16.15.69, 00:48:55, GigabitEthernet0/2/0
C    209.100.12.0/24 is variably subnetted, 2 subnets, 2 masks
L    209.100.12.0/29 is directly connected, GigabitEthernet0/0
L    209.100.12.2/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 is directly connected, GigabitEthernet0/0

RB#show ip route st
RB#show ip route static
S*   0.0.0.0/0 is directly connected, GigabitEthernet0/0

RB#
```

```
MS-D22#show ip route
[110/3] via 172.16.14.194, 00:47:07, Vlan60
[110/3] via 172.16.14.226, 00:47:07, Vlan65
[110/3] via 172.16.15.2, 00:47:07, Vlan70
[110/3] via 172.16.15.34, 00:47:07, Vlan75
O    172.16.15.68 [110/2] via 172.16.15.82, 00:47:44, GigabitEthernet1/1/1
O    172.16.15.72 [110/2] via 172.16.15.82, 00:47:07, GigabitEthernet1/1/1
O    172.16.15.76 [110/2] via 172.16.0.2, 00:47:07, Vlan10
[110/2] via 172.16.8.2, 00:47:07, Vlan15
[110/2] via 172.16.10.2, 00:47:07, Vlan20
[110/2] via 172.16.11.2, 00:47:07, Vlan25
[110/2] via 172.16.12.2, 00:47:07, Vlan30
[110/2] via 172.16.13.2, 00:47:07, Vlan35
[110/2] via 172.16.14.2, 00:47:07, Vlan40
[110/2] via 172.16.14.66, 00:47:07, Vlan45
[110/2] via 172.16.14.162, 00:47:07, Vlan55
[110/2] via 172.16.14.194, 00:47:07, Vlan60
[110/2] via 172.16.14.226, 00:47:07, Vlan65
[110/2] via 172.16.15.2, 00:47:07, Vlan70
[110/2] via 172.16.15.34, 00:47:07, Vlan75
O*E2 0.0.0.0/0 [110/1] via 172.16.15.82, 00:47:44, GigabitEthernet1/1/1
```

Para visualizar las configuraciones completas acceda a la carpeta configuraciones.

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